

- TR 3.1
- Corollary 4.5



- Lemma 2.8
- Remark 3.2



= Sect 5. (Maybe)

TR 2.1

(weak H-T $\rightarrow$ π -monotone emb.)

G has
an π -monot.
 Γ (all edges
are even).



G has an
 π -monotone emb.,
with $\rightarrow$ vertex locations
 $\rightarrow$ rotation system

* proof by induction:

- base case: $|V(G)| = 1 \Rightarrow \pi$ -monotone emb.

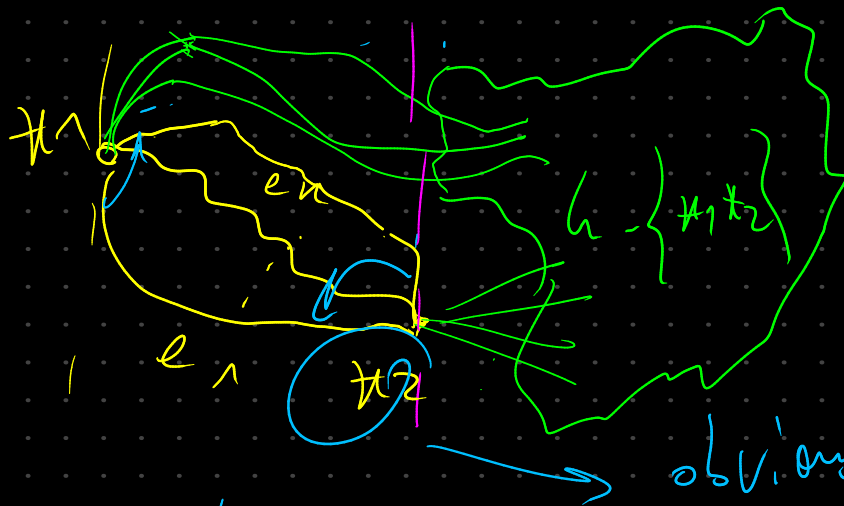
if G consists of multiple components



$\Rightarrow$ we can assume that G is connected.

< induction:

* case 1



Rot_{x_2} has consecutive edges $e_i)_{i=1}^k$

obvious that all the edges on the left comes from x_1

how about Rot_{x_1} ?

let's consider edge $f = x_1 x_l, l > 2$

so f falls btw e_i, e_j in rotation at x_1

if that is true, we will have odd cross between f and either i or j .



it's hard and it needs to get out of the band is $x(x_2) < x(x_3)$

$\Rightarrow e_1, \dots, e_k$ are consec. in $\text{Rot}_{x_1}, \text{Rot}_{x_2}$,
 and $\downarrow$ order is mirrored at
 Rot_{x_1} and Rot_{x_2} .

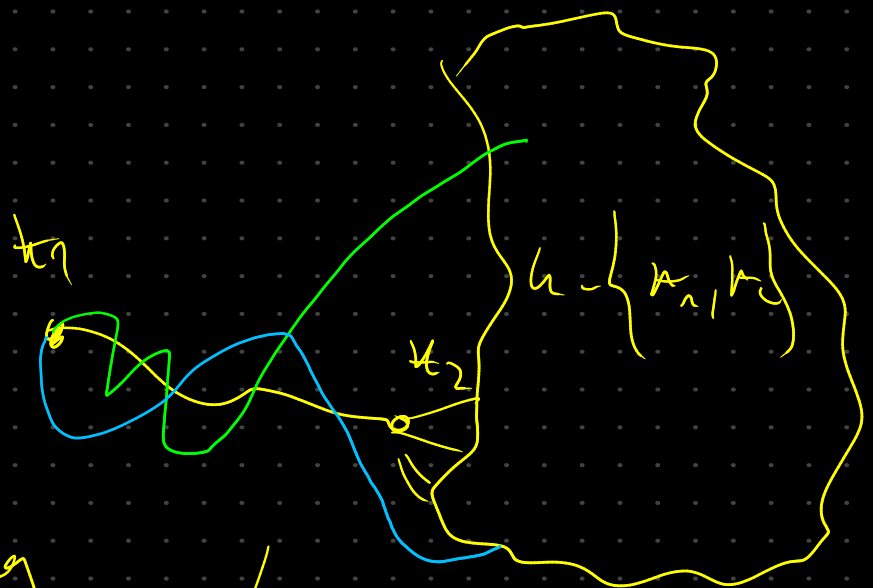


by induction  has the req
 for the embedding

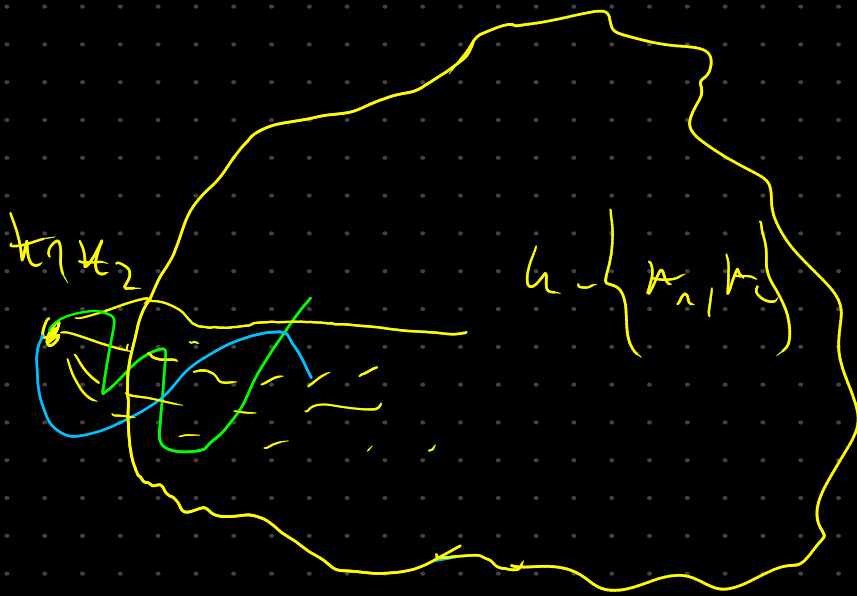


embedding

Case 2



Move x_2 along
 (x_1, x_2)
 till we reach x_1



(x_1, x_2) even (crosses every other edge
an even number of times)

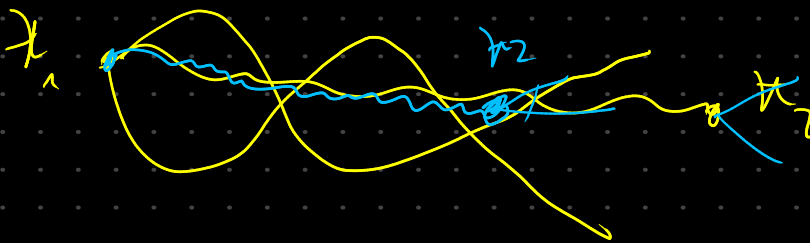
I think it's like shrinking the level
distance, it changes nothing from the h .



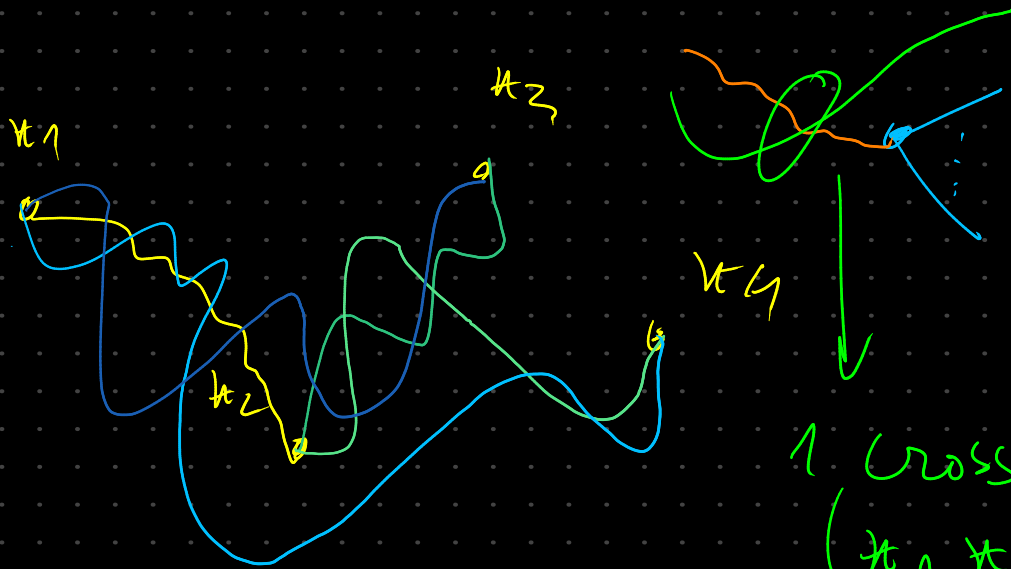
Since x_1, x_2 need to cross
either from above or below
Bot of x_2 will be consecutive

Since x_1, x_2

the cross we remove from e , will
be added to one of the edges of
 x_2 on the right
if we remove
even crosses



We add even crosses to the entirety of
edges x_2, x_4



1 cross of
(x_1, x_2)
will cross one time
all the edges
 x_2, x_4 in $E(V)$

if (x_1, x_2) an even edge (even crossing)

$\Rightarrow$ cross eventimes (x_2, x_4)